

Review retrospective:
*the universal-scaling identity — an exact transport, checked
four ways*

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Abstract

A soundness review of `Laplace/OneD/NearDegenerate.lean`: the exact universal-scaling identity $\langle x^{2n} \rangle_{L_\varepsilon^{nd}, t} = \varepsilon^n \langle y^{2n} \rangle_{V, \varepsilon^2 t}$ relating the near-degenerate potential $L_\varepsilon^{nd}(x) = \frac{\varepsilon}{2}x^2 + \frac{x^4}{24}$ to the universal $V(y) = \frac{y^2}{2} + \frac{y^4}{24}$ via $x = \sqrt{\varepsilon}y$. Result: **a clean fidelity pass with one dead-open removal** — the substantive math (an exact change-of-variables transport) checked out on all four load-bearing steps.

1 Setting

Unlike the templated `TwoD/` lift files, this is a genuine change-of-variables result: the substitution $x = \sqrt{\varepsilon}y$ sends tL_ε^{nd} to $\varepsilon^2 tV$ *exactly*, so every near-degenerate Gibbs moment factors through the universal posterior at effective inverse temperature $T = \varepsilon^2 t$. It is the formal core of the near-degenerate-crossover note.

2 What was reviewed

Two potential abbreviations, an algebraic core lemma, and four theorems: the integral substitution ($\int f e^{-tL_\varepsilon^{nd}} = \sqrt{\varepsilon} \int f(\sqrt{\varepsilon} \cdot) e^{-\varepsilon^2 tV}$), the partition scaling ($Z_{L_\varepsilon^{nd}} = \sqrt{\varepsilon} Z_V(\varepsilon^2 t)$), the Gibbs expectation transport, and the headline monomial scaling.

3 Fidelity / soundness findings

Sound; checked four ways. (1) The algebraic core $L_\varepsilon^{nd}(\sqrt{\varepsilon}y) = \varepsilon^2 V(y)$ expands correctly ($\frac{\varepsilon}{2}(\varepsilon y^2) + \frac{\varepsilon^2 y^4}{24} = \varepsilon^2(\frac{y^2}{2} + \frac{y^4}{24})$). (2) The Jacobian is in the right direction: $x = \sqrt{\varepsilon}y \Rightarrow dx = \sqrt{\varepsilon}dy$, matching the $\sqrt{\varepsilon}$ prefactor (the proof rearranges Mathlib’s `integral_comp_mul_right` output). (3) The $\sqrt{\varepsilon}$ prefactors of numerator and partition cancel in the expectation transport. (4) The headline factor is $(\sqrt{\varepsilon})^{2n} = \varepsilon^n$. The reviewer independently confirmed all four.

One report-only scope note: the module docstring’s prose about the $T \ll 1$ and $T \gg 1$ regimes (“governed by the pure-quartic core”, “tracks the standard Laplace asymptotics”) is *contextual* — this file proves the exact transport/scaling identity, not those asymptotic regime statements. Left intact as expository scope-setting, not gutted (the same treatment as other research-context framing).

4 Simplifications applied

One dead open. `open Real` is dead (`Real.sqrt/Real.exp/Real.sq_sqrt/Real.sqrt_pos` all qualified); `open MeasureTheory` stays (bare $\int$ throughout). Module green at 2697 jobs, no warnings, signatures byte-identical. GPT affirmed.

5 Disagreements and open questions

None. The reviewer’s several `show→change` and `simpa`-compression suggestions were declined as proof-golf.

6 Lessons

- **A genuine change-of-variables result rewards a four-point fidelity check.** Unlike a templated repackaging, this file’s correctness hinges on four independently-falsifiable facts (algebraic identity, Jacobian direction, prefactor cancellation, exponent count). Enumerating them — rather than trusting the build — is what a fidelity review adds over a green lake build.
- **Contextual regime prose is expository, not a proof obligation.** A docstring that situates an exact identity within a $T \ll 1/T \gg 1$ crossover story is scope-setting; reviewing it means confirming the file does not *claim* to prove the regimes, not flagging their absence.

7 Follow-ups

- Remaining `OneD/` surface: `UniversalAsymptotics` (the natural companion — the $T \gg 1/T \ll 1$ regime statements this file’s docstring gestures at), `HarmonicCovGlobalConstancy`, `HarmonicPerturbedMoments`, the `Harmonic*` monomials; and the `Multi/Covariance*` trio.
- Proof-golf forward tide: the `show→change` and `simpa` compressions across this file.